

MATHEMATICS

TARGET : JEE- Advanced

CAPS-19**DIFFERENTIAL EQUATION****SCQ (Single Correct Type) :**

- If $(1+x^2)\frac{dy}{dx} = x(1-y)$, $y(0) = \frac{4}{3}$, then the value of $y(\sqrt{8}) + \frac{8}{9}$ is :
(A) 4 (B) 2 (C) 3 (D) 5
- Solution of the differential $y' = \frac{3yx^2}{x^3 + 2y^4}$ is
(A) $x^3y^{-1} = \frac{2}{3}y^3 + c$ (B) $x^2y^{-1} = \frac{2}{3}y^3 + c$ (C) $xy^{-1} = \frac{2}{3}y^3 + c$ (D) None of these
- If $ydx - dy = e^{-x}y^4dy$ and $y = 1$ at $x = -\ln 3$, then at $y = 3$, $x = \ln k$, k is
(A) 21 (B) 24 (C) 27 (D) 81
- The solution of the differential equation $(x \cot y + \ln(\cos x))dy + (\ln(\sin y) - y \tan x)dx = 0$ is _____.
(A) $(\sin x)^y (\cos y)^x = c$ (B) $(\sin y)^x (\cos x)^y = c$
(C) $(\sin x)^y (\cos y)^x = c$ (D) $(\cot x)^y (\cot y)^x = c$
- If x and y are related, Also $\frac{d^2x}{dy^2} \cdot \left(\frac{dy}{dx}\right)^3 + k \frac{d^2y}{dx^2} = 0$. Then value of k is _____.
(A) -1 (B) 1 (C) -2 (D) 2
- Solution of the differential equation $y(xy + 2x^2y^2)dx + x(xy - x^2y^2)dy = 0$ is given by
(A) $2\log|x| - \log|y| - \frac{1}{xy} = C$ (B) $2\log|y| - \log|x| - \frac{1}{xy} = C$
(C) $2\log|x| + \log|y| + \frac{1}{xy} = C$ (D) $2\log|y| + \log|x| + \frac{1}{xy} = C$
- The solution of the differential equation $\frac{dy}{dx} = \frac{1}{xy(x^2 \sin y^2 + 1)}$ is, ('c' being an arbitrary constant).
(A) $x^2(\cos y^2 - \sin y^2 - 2ce^{-y^2}) = 2$ (B) $x^2(\cos y^2 - \sin y^2 - 2ce^{-y^2}) = 4c$
(C) $y^2(\cos x^2 - \sin x^2 - 2ce^{-y^2}) = 2$ (D) $y^2(\cos x^2 - \sin x^2 - ce^{-y^2}) = 4c$

MCQ (One or more than one correct) :

- The solution of $\left(\frac{dy}{dx}\right)^2 - 2\left(x + \frac{1}{4x}\right)\frac{dy}{dx} + 1 = 0$
(A) $y = x^2 + c$ (B) $y = \frac{1}{2}\ln(x) + c, x > 0$ (C) $y = \frac{x}{2} + c$ (D) $y = \frac{x^2}{2} + c$

9. A function $y = f(x)$ satisfying the differential equation $\frac{dy}{dx} \cdot \sin x - y \cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y \rightarrow 0$ as $x \rightarrow \infty$ then the statement which is correct is
- (A) $\lim_{x \rightarrow 0} f(x) = 1$ (B) $\int_0^{\pi/2} f(x) dx$ is less than $\frac{\pi}{2}$
- (C) $\int_0^{\pi/2} f(x) dx$ is greater than unity (D) $f(x)$ is an odd function
10. Let T be the triangle with vertices $(0, 0)$, $(0, c^2)$ and (c, c^2) and let R be the region between $y = cx$ and $y = x^2$ where $c > 0$ then
- (A) $\text{Area}(R) = \frac{c^3}{6}$ (B) $\text{Area of } R = \frac{c^3}{3}$
- (C) $\lim_{c \rightarrow 0^+} \frac{\text{Area}(T)}{\text{Area}(R)} = 3$ (D) $\lim_{c \rightarrow 0^+} \frac{\text{Area}(T)}{\text{Area}(R)} = \frac{3}{2}$

Comprehension Type Question:

Comprehension # 1

Suppose $f(x)$ and $g(x)$ are differentiable functions such that $x g(f(x)) \cdot f'(g(x)) \cdot g'(x) = f(g(x)) \cdot g'(f(x))$

$f'(x) \forall x \in \mathbb{R}$ and $f(x)$ & $g(x)$ are positive for all $x \in \mathbb{R}$. Also $\int_0^x f(g(t)) dt = \frac{1}{2}(1 - e^{-2x}) \forall x \in \mathbb{R}$, $g(f(0))$

$= 1$ and $h(x) = \frac{g(f(x))}{f(g(x))} \forall x \in \mathbb{R}$.

11. The graph of $y = h(x)$ is symmetric with respect to the line:
- (A) $x = -1$ (B) $x = 0$ (C) $x = 1$ (D) $x = 2$
12. The value of $f(g(0)) + g(f(0))$ is equal to:
- (A) 1 (B) 2 (C) 3 (D) 4
13. The largest possible value of $h(x) \forall x \in \mathbb{R}$ is
- (A) 1 (B) $e^{1/3}$ (C) e (D) e^2

Comprehension # 2

Let $y = f(x)$ satisfy the equation $f(x) = (e^{-x} + e^x) \cos x - 2x - \int_0^x (x-t)f'(t) dt$. Then,

14. $f(x)$ satisfies the differential equation
- (A) $\frac{dy}{dx} + y = e^x (\cos x - \sin x) - e^{-x} (\cos x + \sin x)$
- (B) $\frac{dy}{dx} = y + e^x (\cos x + \sin x) + e^{-x} (\cos x - \sin x)$
- (C) $\frac{dy}{dx} + y = e^x (\cos x + \sin x) - e^{-x} (\cos x + \sin x)$
- (D) None of these

15. $f'(0) + f''(0) =$
 (A) -1 (B) 2 (C) 0 (D) 1
16. $f(x)$ as a function of x equals
 (A) $e^{-x} (\cos x - \sin x) + \frac{e^x}{5} (3\cos x + \sin x) + \frac{2}{5} e^{-x}$
 (B) $e^{-x} (\cos x + \sin x) + \frac{e^x}{5} (3\cos x - \sin x) - \frac{2}{5} e^{-x}$
 (C) $e^{-x} (\cos x - \sin x) + \frac{e^x}{5} (3\cos x - \sin x) + \frac{2}{5} e^{-x}$
 (D) None of these

Numerical based Questions :

17. If $ye^y dx = (y^3 + 2xe^y) dy$, $y(0) = 1$, then the value of x when $y = 1$ is
18. If particular solution of $p + \cos px \sin y = \sin px \cos y$ where $\left(p = \frac{dy}{dx}\right)$ is
 $y = a\sqrt{x^2 - b} - \sin^{-1} \frac{\sqrt{x^2 - c}}{x}$ then evaluate $\left| \frac{b+c}{a} \right|$
19. The solution of $x^2 dy - y^2 dx + xy^2(x-y) dy = 0$ is $\ln \left| \frac{1}{x} - \frac{1}{y} \right| = \frac{y^2}{k} + C$, then the value of k is _____.
20. Suppose a solution of the differential equation $(xy^3 + x^2y^7) \frac{dy}{dx} = 1$ satisfies the initial condition $y\left(\frac{1}{4}\right) = 1$. Then the value of $-\frac{5}{4} \times \frac{dy}{dx}$ when $y = -1$, is _____.
21. Let $y = f(x)$ be a curve C_1 passing through $(2,2)$ and $\left(8, \frac{1}{2}\right)$ and satisfying a differential equation $y\left(\frac{d^2y}{dx^2}\right) = 2\left(\frac{dy}{dx}\right)^2$. Curve C_2 is the director circle of the circle $x^2 + y^2 = 2$. If the shortest distance between the curves C_1 and C_2 is $(\sqrt{p} - q)$ where $p, q \in \mathbb{N}$, then find the value of $(p^2 - q)$.

Matrix Match Type :

22. Match the following

Column-I	Column-II
(A) Solution of the differential equation $y'' + e^{2y}(y')^3 = 0$ is	(p) $x = \frac{1}{4}e^{2y} + c_1y + c_2$
(B) Solution of the differential equation $2x(x+y)y' = 3y^2 + 4xy$ at $y(1) = 1$ is	(q) $y^2 + 2xy = 3x^3$
(C) Equation of orthogonal trajectories of the curve $y = \tan x + k$ is	(r) $2x + 4y + \sin 2x = c$
(D) Solution of the differential equation $y' + 4xy + xy^3 = 0$ is	(s) $y^2 = \left(ce^{4x^2} - \frac{1}{4}\right)^{-1}$

- (A) A-p; B-q; C-r; D-s (B) A-s; B-p; C-r; D-q
(C) A-p; B-q; C-s; D-r (D) A-p; B-s; C-q; D-r
- 23. Column I Column II**
- (A) If the function $y = e^{4x} + 2e^{-x}$ is a solution of the differential equation $\frac{\frac{d^3y}{dx^3} - 13 \frac{dy}{dx}}{y} = K$ then the value of $K/3$ is (P) 3
- (B) Number of straight lines which satisfy the differential equation $\frac{dy}{dx} + x \left(\frac{dy}{dx} \right)^2 - y = 0$ is (Q) 4
- (C) If real value of m for which the substitution, $y = u^m$ will transform the differential equation, $2x^4y \frac{dy}{dx} + y^4 = 4x^6$ into a homogenous equation then the value of $2m$ is (R) 2
- (D) If the solution of differential equation $x^2 \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} = 12y$ is $y = Ax^m + Bx^{-n}$ then $|m + n|$ can be (S) 7
- (A) A(R); B(Q); C(S); D(P) (B) A(P); B(S); C(Q); D(R)
(C) A(Q); B(R); C(P); D(S) (D) A(S); B(P); C(Q); D(R)

Subjective based Questions :

- 24.** Let the curve $y = f(x)$ passes through $(4, -2)$ satisfy the differential equation,
 $y(x + y^3) dx = x(y^3 - x) dy$ & let $y = g(x) = \int_{1/8}^{\sin^2 x} \sin^{-1} \sqrt{t} dt + \int_{1/8}^{\cos^2 x} \cos^{-1} \sqrt{t} dt$,
 $0 \leq x \leq \frac{\pi}{2}$. Then find the area of the region bounded by curves $y = f(x)$, $y = g(x)$ and $x = 0$.
- 25.** Solve the equation $x \int_0^x y(t) dt = (x + 1) \int_0^x t y(t) dt$, $x > 0$

ANSWER KEY OF CAPS-19

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|----------|---------|---------|--|----------------------------------|
| 1. (B) | 2. (A) | 3. (C) | 4. (B) | 5. (B) |
| 6. (A) | 7. (A) | 8. (AB) | 9. (ABC) | 10. (AC) |
| 11. (C) | 12. (B) | 13. (C) | 14. (A) | 15. (C) |
| 16. (C) | 17. (0) | 18. (2) | 19. (2) | 20. (4) |
| 21. (62) | 22. (A) | 23. (C) | 24. $\frac{1}{8} \left(\frac{3\pi}{16} \right)^4$ | 25. $y = \frac{c}{x^3} e^{-1/x}$ |